Project

Second Year Project Presentation

College FAQ Chatbot using NLP

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Introduction to Chatbots

What is a Chatbot?
A chatbot is an AI-powered software that simulates human conversation through text or voice.
Uses of Chatbots:
- Customer service and support
- Education and student assistance
- Daily life applications

Key Benefits:
- 24/7 availability
- Quick, instant responses
- Enhanced user engagement

Project Objective

- Purpose: Build an intelligent chatbot to answer college FAQs instantly
- Problem: Students often struggle to find quick, accurate answers to common questions
- Solution: NLP-powered chatbot providing instant, reliable responses 24/7
- Goal: Improve student experience and reduce administrative workload

Technologies Used

Python

Core programming language for building the chatbot system.

Tkinter

Python library for creating the graphical user interface.

NLTK

Natural Language Toolkit for NLP processing tasks.

Integration

All components work together seamlessly.

What is NLP?

Natural Language Processing (NLP) enables computers to understand and interpret human language.

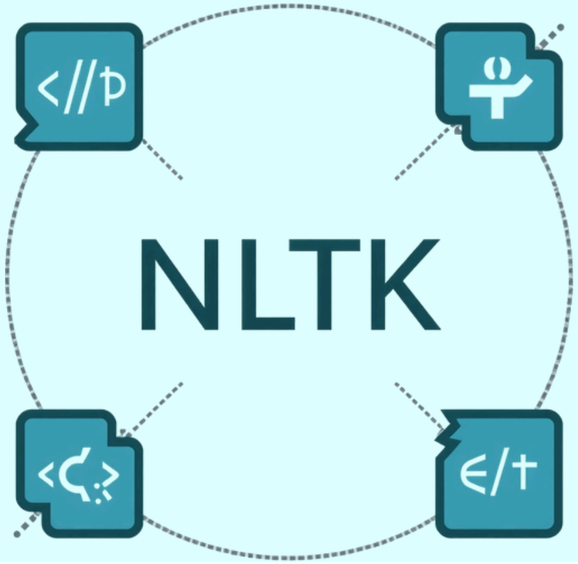
•Helps chatbot understand user queries

•Converts text into meaningful data

•Key techniques: tokenization, intent recognition

•Makes human-computer interaction natural

What is NLTK?



NLTK (Natural Language Toolkit) is a powerful Python library for natural language processing. Key features include tokenization, stemming, and parsing. It provides essential tools for building intelligent chatbots that can understand and process human language effectively.

System Working

User Input

Student types question into chatbot interface.

NLP Processing

NLTK tokenizes and analyzes the text.

Intent Matching

System matches query to FAQ database.

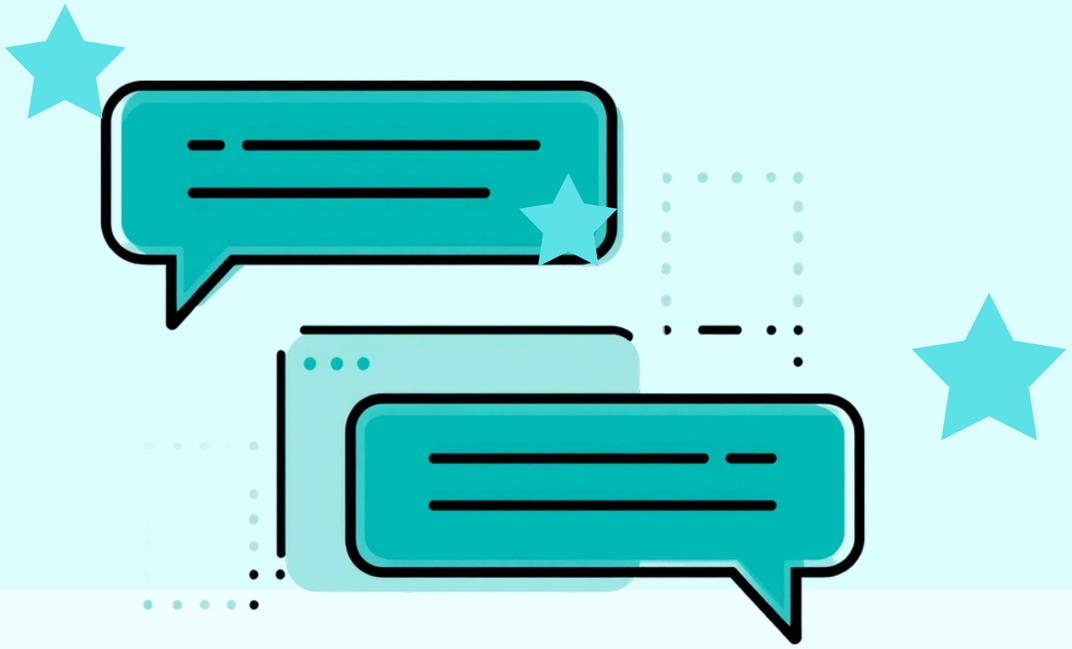
Response Generation

Chatbot delivers accurate answer to user.

Implementation

- Python code integrates NLTK for text processing and intent recognition
- Tkinter GUI provides user-friendly chat interface
- Response system matches user queries to FAQ database
- Pattern matching identifies question intent and retrieves answers

Output Screenshots



The chatbot interface displays a clean conversation window where students can type their questions. Sample interactions show queries like "What are the library hours?" and "How do I apply for hostel?" with instant, accurate responses. The GUI provides a user-friendly experience with clear message bubbles and readable text.

Sample User Queries

College FAQ Chatbot

Bot: Welcome to the College FAQ Chatbot! How can I assist you today?
You: Hello
Bot is typing...
Bot: Hello! Welcome to our college chatbot.
You: What is the admission process?
Bot is typing...
Bot: You can apply online through the college website.
You: What courses are offered?
Bot is typing...
Bot: The college offers programs in Civil, Mechanical, Computer Science, Electronics, and Management. Please specify if you want details about a particular course.
You: What is the fee structure?
Bot is typing...
Bot: The fee is ₹1,50,000 per year for B.Tech course. Please check for the other courses from the college website.

College FAQ Chatbot

Bot: Welcome to the College FAQ Chatbot! How can I assist you today?
You: Hello
Bot is typing...
Bot: Hello! Welcome to our college chatbot.
You: Is hostel available?
Bot is typing...
Bot: Yes, hostel facilities are available.
You: What are the college timings?
Bot is typing...
Bot: College runs from 8 AM to 3 PM.
You: What about placements?
Bot is typing...
Bot: We provide good placement opportunities.
You: What departments are available?
Bot is typing...
Bot: Our college offers departments like Computer Science, AI & Data Science, Electronics, Mechanical, Civil, and Management.
You: Are there sports facilities?
Bot is typing...
Bot: Sports facilities include cricket, football, and indoor games.
You: What events are organized in college?
Bot is typing...
Bot: We organize cultural fests, tech fests, and seminars.

College FAQ Chatbot

Bot: Welcome to the College FAQ Chatbot! How can I assist you today?
You: Hello
Bot is typing...
Bot: Hello! Welcome to our college chatbot.
You: What is the contact information?
Bot is typing...
Bot: You can contact us at info@college.com or call 1234567890.
You: Where is the college located?
Bot is typing...
Bot: The college is located in Kolkata, West Bengal.
You: Is WiFi available in the college?
Bot is typing...
Bot: Yes, WiFi is available across the campus.

Key Features of the Chatbot

GUI-Based

Intuitive graphical interface for easy interaction.

Fast Response

Quick answers with minimal wait time.

FAQ Coverage

Covers common college FAQs effectively.

User-Friendly

Simple design accessible for all users.

College FAQ Chatbot

Thank You!

NLP-powered chatbot helps students get quick answers. Future: voice interface, AI enhancements, web app.